

Course-Based Projects

- Students can work individually or in groups of 2/3/4. The complexity of the project increases as the number of team members increases. You cannot change the project or group after approval.
- Deadlines:
 - a. Oct 9, 2026 @ 5 PM - Abstract approval. Email me. No need to meet me.
 - b. Oct 26, 2026 - Report submission.
- You have the opportunity to **BE CREATIVE** and decide on the projects using ANY (after approval) equipment in the SA lab. Yes - You can use ChatGPT. Take videos and put them on YouTube.
- If interested, you are allowed to work on Experiment 6 & Experiment 8. You are also allowed to perform additional work on one of the experiments in the SA Lab.
- Equipment List:
 - a. Microcontrollers: Arduino UNO R4 WiFi/Raspberry Pi 5
 - b. Sensors: Linear Potentiometer/IMU (6/9-axis)/Ultrasonic/IR/Flex/Current /Environmental/ Color/Optical Encoder/Strain Gauge Load Cell/LiDAR/RGB-D Camera/Linear Solenoid/Global & Rolling Shutter.
 - c. Actuators: DC/BLDC/Stepper/Haptic Vibration Motor
 - d. Pan-Tilt Assembly (For Experiment 12).
 - e. Depending on the project, feel free to buy any other equipment if required.
- Submission Guidelines:
 - a. Abstract: (Maximum 1 page. Latex IEEE format - Talk to me for specifics)
 - Project Title, Student Name/s, Roll Numbers
 - Describe the problem that your project will solve
 - List of equipment and how will the system operate?
 - Expected result.
 - For group projects: Give each student a specific task.
 - b. Report (Maximum 4 pages. Latex IEEE format - Talk to me for specifics)
 - Abstract
 - Introduction - Some background information
 - Methodology - Diagrams
 - Experimental Results - Plots/Code/Tables/Videos/Pictures
 - Analysis of Results
 - Conclusion
 - Acknowledgements - Describe each student's work in the report.
 - References

Internal Practical Examination

- Dates:
 - a. Oct 12, 2026 - Monday batch (10:00 AM to 12:00 PM)
 - b. Oct 22, 2026 - Thursday batch (1:40 PM to 3:40 PM)
- Individual examination. Closed book exam - phones/notes/cheat-sheets are NOT allowed.
- Duration: 90 minutes
- Maximum mark: 10
- Each student will randomly draw an experiment. Maximum of two changes are allowed - you will lose 1 mark for each change. Decide change immediately.
- You will have to write small functions. Understand the logic of each experiment - which pins are connected and why? What are the operations to be performed variables (Think back to IMU)?
- Rubric:
 - a. Initial Procedure and Write-Up - 1.5 marks
 - Component identification (0.5)
 - Aim (0.5)
 - Implementation Plan (0.5)
 - b. Conduct of Experiment/Program - 3.5 marks
 - Setup/Implementation (1)
 - Code (1.5)
 - Operation & Demonstration (1) - Ask the examiner to check the result.
 - c. Analysis and Evaluation - 3 marks
 - Observations (1)
 - Analysis (1)
 - Conclusion (1)
 - d. Viva - 2 marks
- List of Experiments:
 - a. Process quadrature encoder data from a live sensor to estimate the displacement and velocity.
 - b. Characterize noise and bias from an IMU stream to estimate stable orientation via digital filtering.
 - c. Analyze live Ultrasonic distance data to model sensor accuracy and identify limitations.
 - d. Control a single DC motor using Arduino and L298N.
 - e. Control two DC motors using Arduino and L298N.
 - f. Control a BLDC motor with an electronic speed controller and Arduino.
 - g. Control two BLDC motors with an electronic speed controller and Arduino.
 - h. Use basic stepper motor control to control a stepper motor using Arduino and L298N.